LAPORAN BUKTI PENGUJIAN APLIKASI KELOMPOK 7 INVENTARIS BARANG

Nama :

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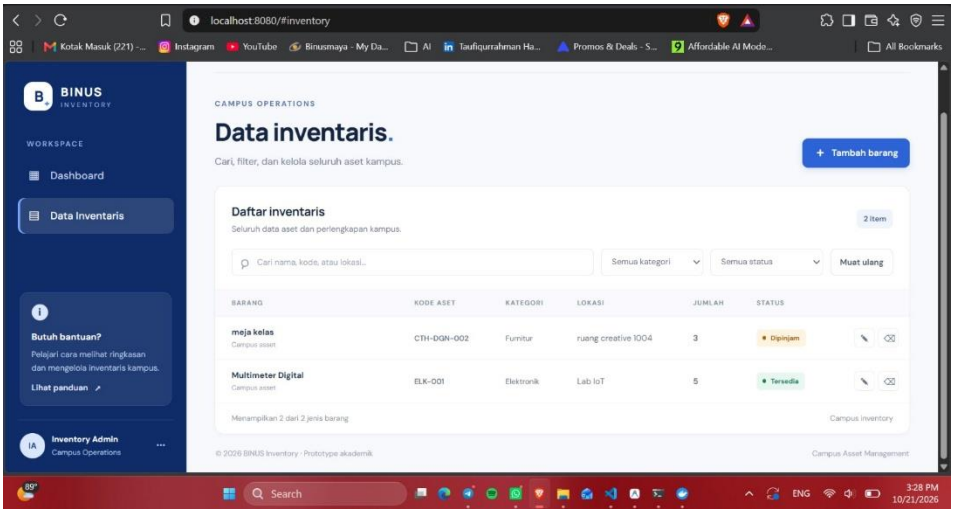
- TAUFIQURRAHMAN HAMDAN AL AYUBI - 2802597450

Bagian 1 : smoke test

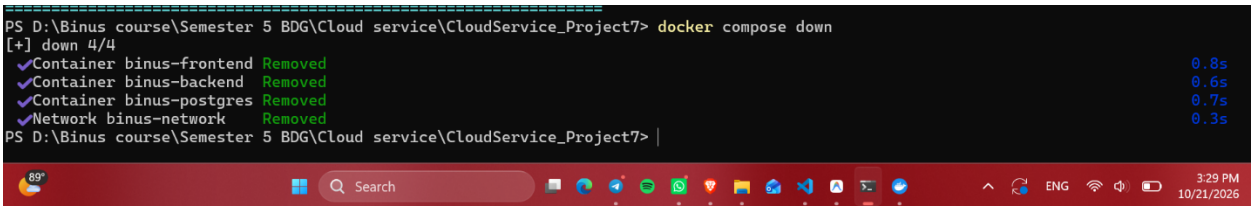
```
PS D:\Binus course\Semester 5 BDG\Cloud service\CloudService_Project7> .\scripts/smoke-test.ps1
================================================================
 MEMULAI AUTOMATED SMOKE TEST: http://127.0.0.1:8080
================================================================
 [PASS] 1. Nginx Reverse Proxy Gateway (/health)
 [PASS] 2. Frontend UI Availability (/)
 [PASS] 3. Backend API & PostgreSQL Health (/api/health)
 [PASS] 4. GET /api/items (Daftar inventaris)
 [PASS] 5. Strict Input Validation (Berhasil tolak data invalid dengan HTTP 400)
 [PASS] 6. POST /api/items (Tambah barang baru valid - HTTP 201)
 [PASS] 7. PUT /api/items/6 (Perbarui barang - HTTP 200)
 [PASS] 8. DELETE /api/items/6 (Hapus barang - HTTP 204)
================================================================
 HASIL SMOKE TEST: Sukses: 8 | Gagal: 0
================================================================
PS D:\Binus course\Semester 5 BDG\Cloud service\CloudService_Project7>
```

Pengujian otomatis menggunakan skrip PowerShell menguji 8 skenario: Nginx Gateway, rute UI, koneksi database PostgreSQL, penolakan payload tidak valid (HTTP 400), dan siklus penuh operasi CRUD.

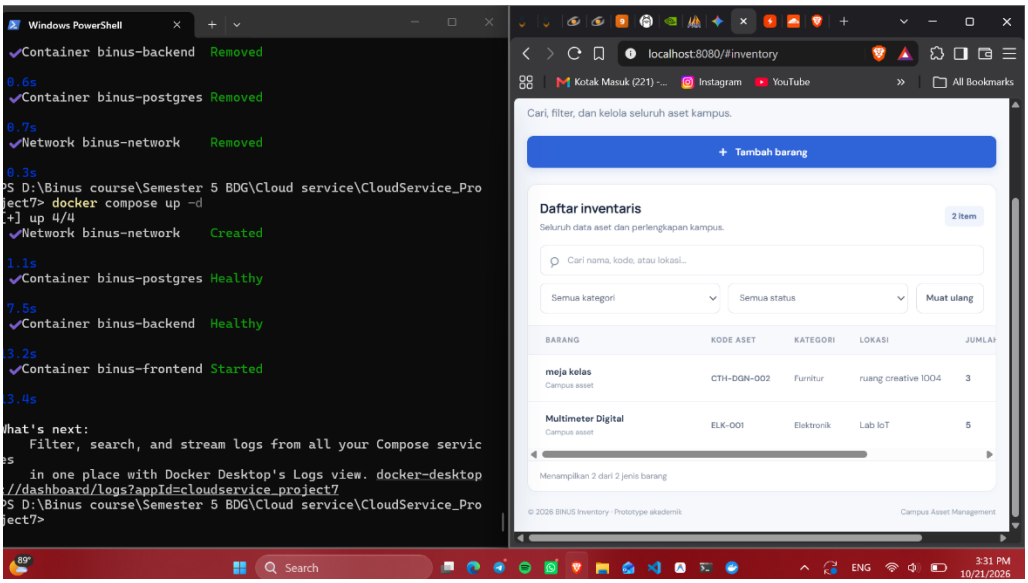
Bagian 2 : uji persistensi data (down => up).



Add item



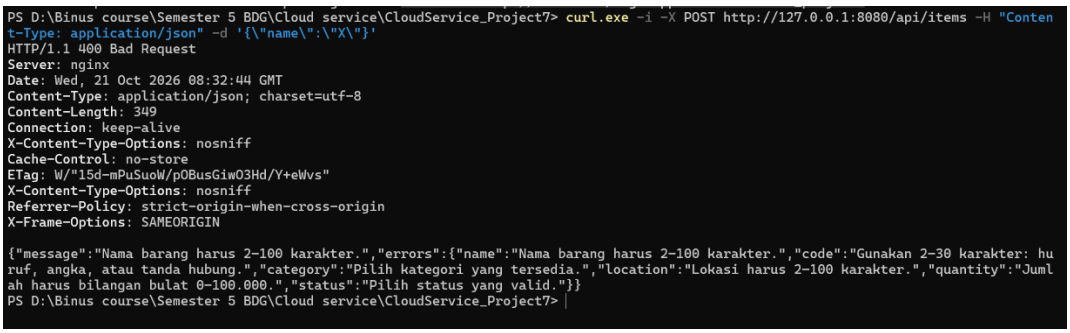
Docker compose down : container dimatikan



Docker compose up -d : container Kembali dinyalakan dan data barang masih ada.

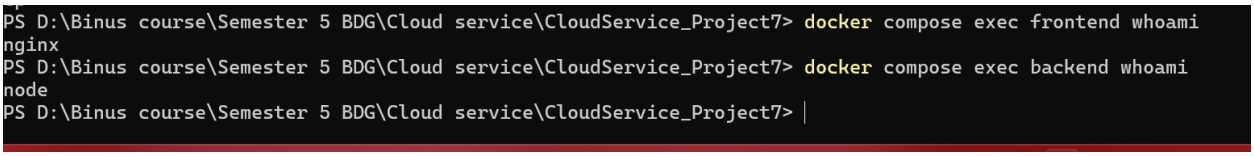
Pengujian membuktikan persistensi data menggunakan Docker Named Volume binus_postgres_data. Data tetap utuh dan tersimpan setelah container di-destroy dan dijalankan ulang.

Bagian 3 : validasi input yang tidak valid (http 400).



Backend menerapkan validasi ketat. Request dengan payload yang tidak memenuhi schema aturan ditolak dengan status HTTP 400 tanpa menyebabkan server error.

Bagian 4: Bukti Keamanan Non-Root User & Healthcheck Container.



Semua service diorkestrasi menggunakan Docker Compose dengan healthcheck otomatis untuk memastikan dependensi database dan backend siap sebelum gateway melayani request.